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College: *Bicol University - College of Science*
Program & Block: *Bachelor of Science in Information Technology - 3A*
Course: *IT 106 - Application Development & Emerging Technology*
Topic: *Group REST API Enhancement and Multi-Client Implementation*

System Process Documentation: ZenFit REST API System

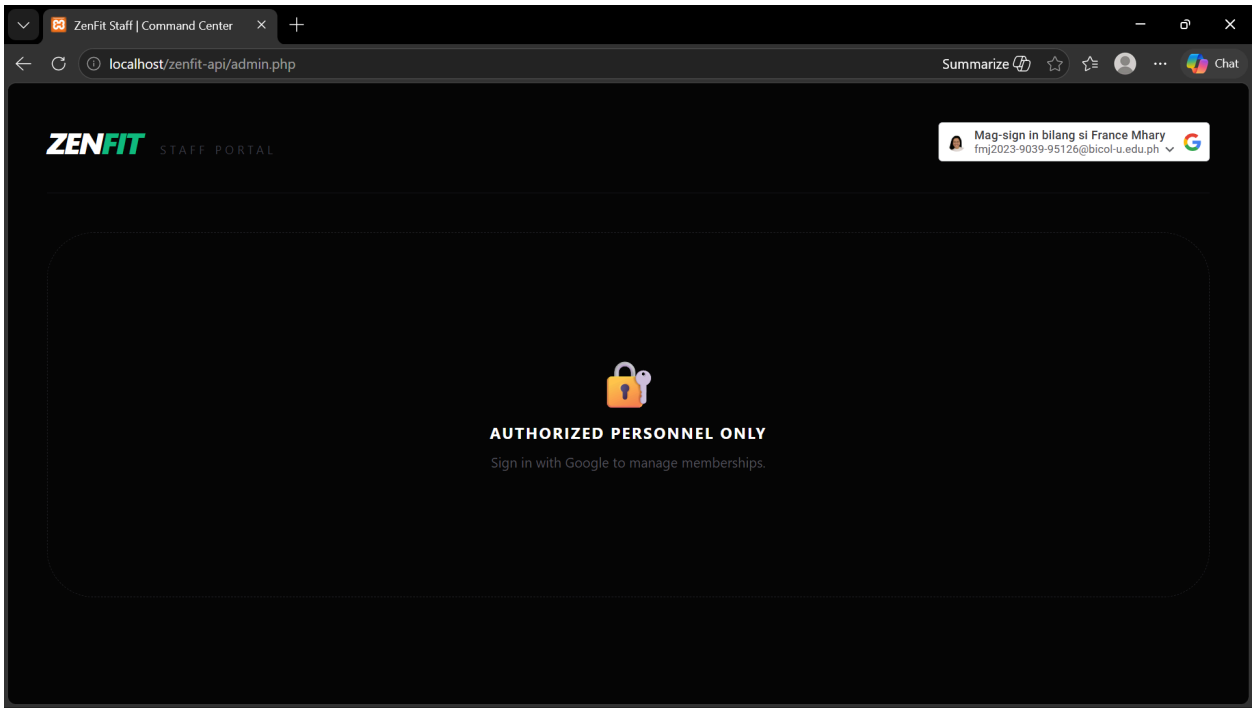

Figure 1. *Access Control & Authentication*

The entry point of the Staff Portal. To ensure security, the system implements Google OAuth 2.0. Access to the enrollment dashboard and the database is

restricted; users must authenticate using an authorized Google account before the REST API functions are exposed in the UI.

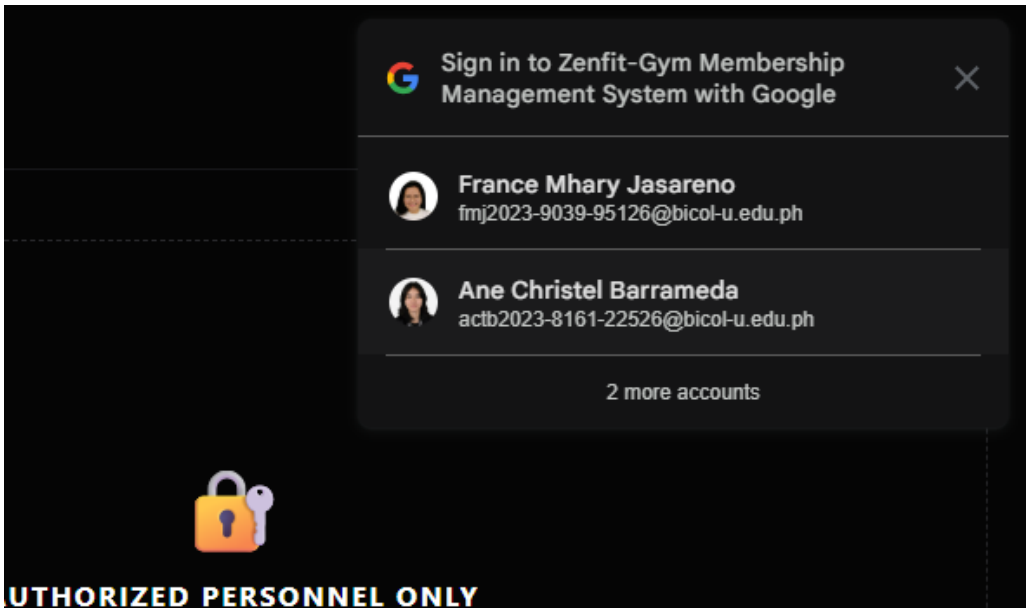

Figure 2: *Account Selection Prompt*

The system initiates the secure login process by displaying a Google account chooser, allowing authorized staff members to select their email to gain access to the ZenFit management tools.

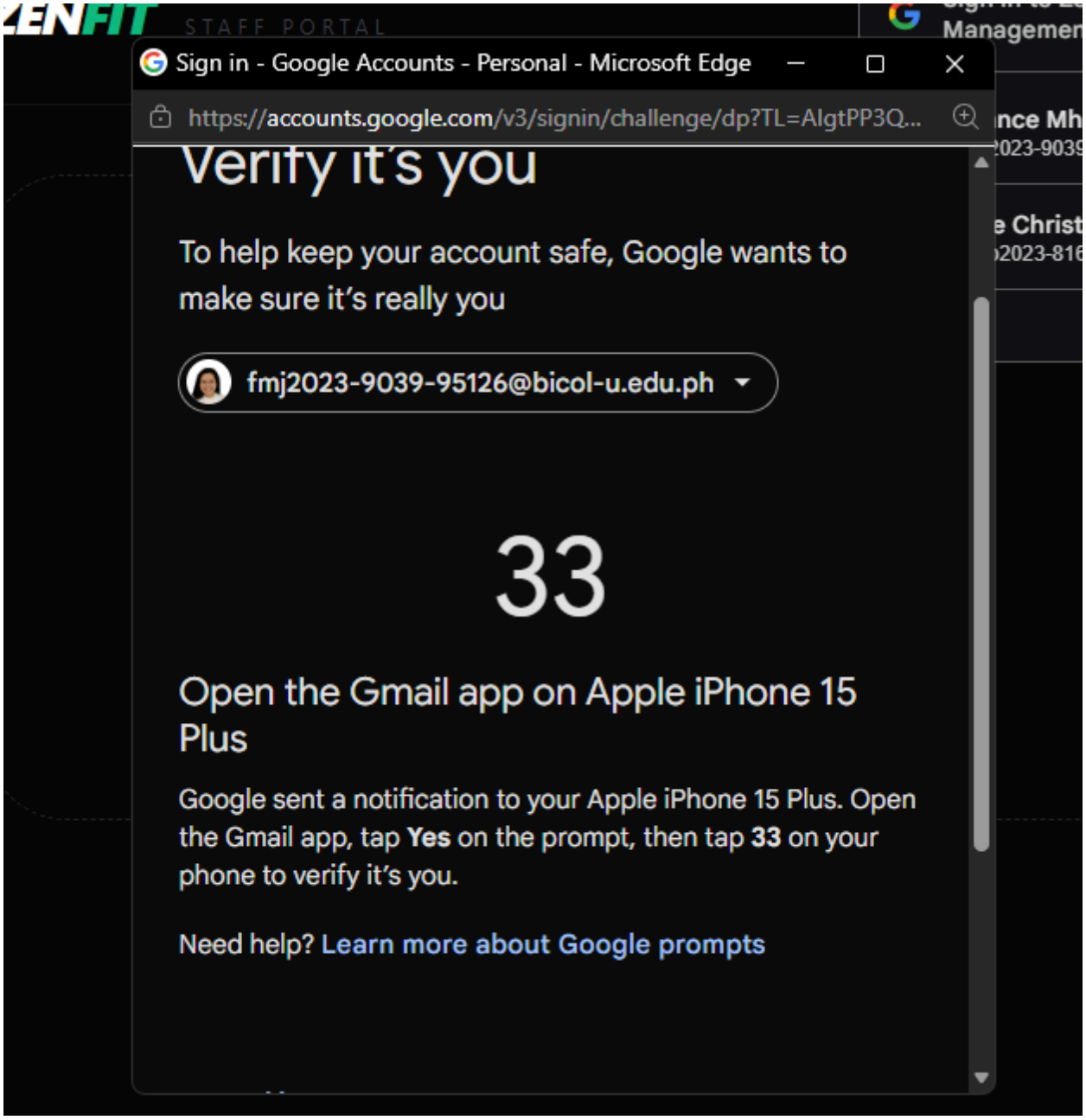

Figure 3. *Verification Code Synchronization*

To enhance system security, a "Verify it's you" challenge is issued, requiring the staff member to complete a mobile push notification check before the administrative dashboard is unlocked.

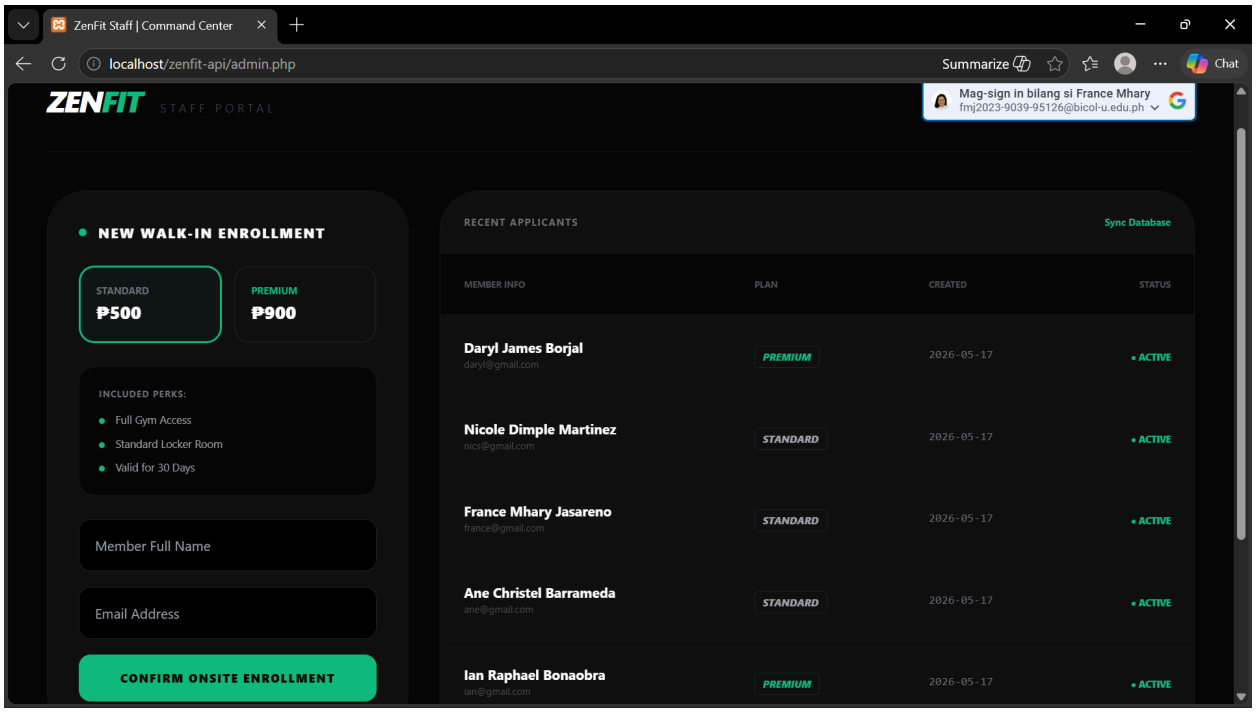

Figure 4. *The Staff Command Center Dashboard*

Once authenticated, the dashboard provides a dual-pane interface. On the left is the Client Enrollment Form which captures member data. On the right is a live view of the Recent Applicants.

Testing:

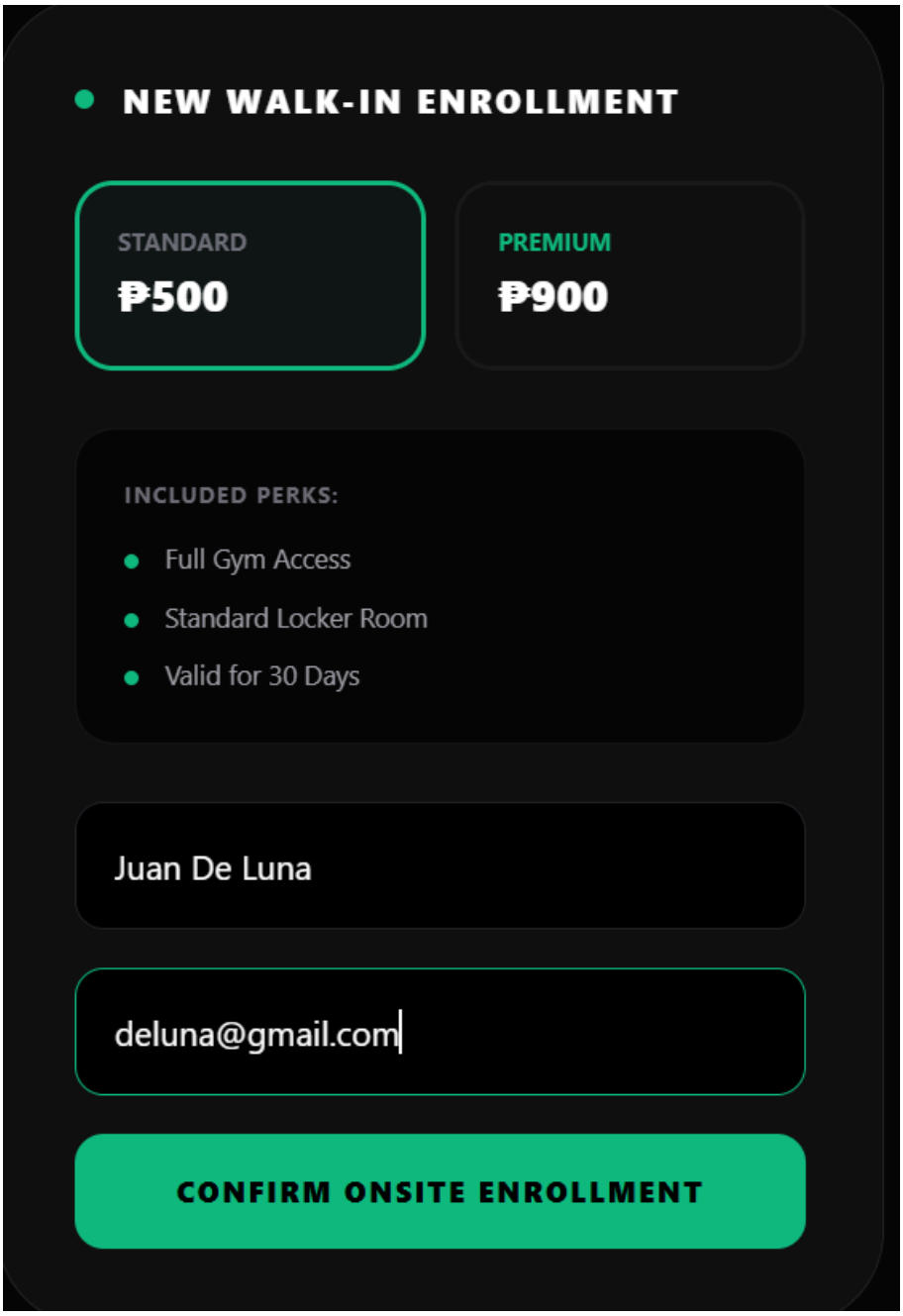

Figure 5. *Data Input & Validation*

This illustrates the client-side data entry process. The system accepts two primary parameters—Member Full Name and Email Address—alongside a selected membership plan (Standard or Premium).

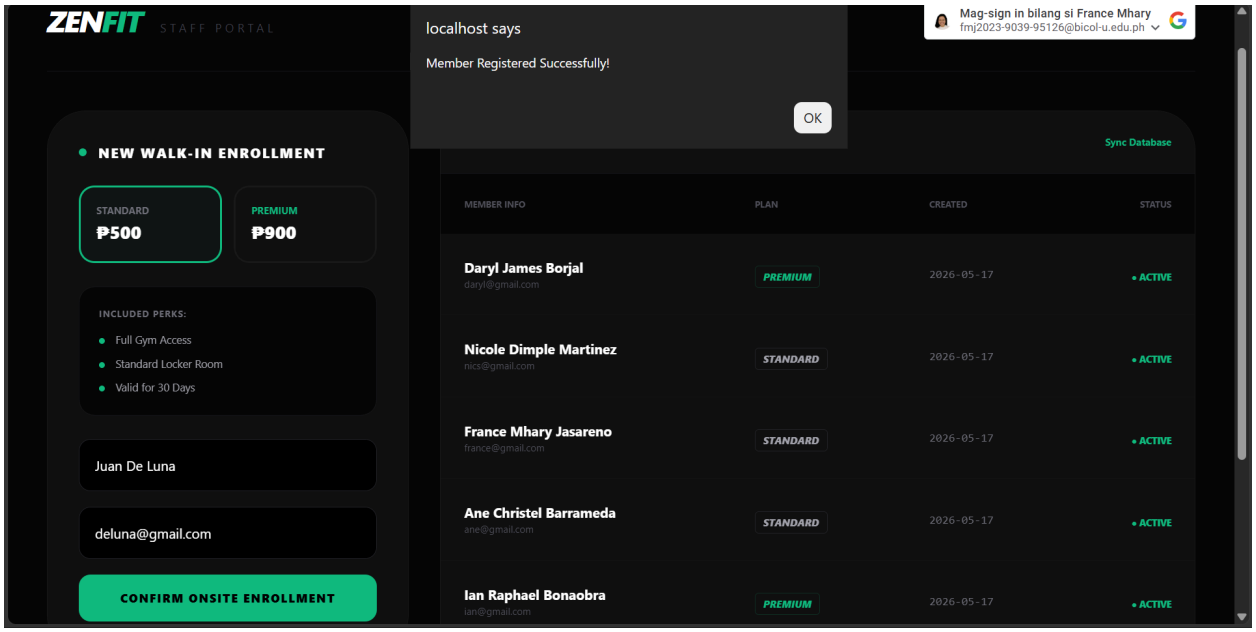

Figure 6: *API Request & Success Response*

The **"Member Registered Successfully!"** alert is triggered by the client-side JavaScript once it receives a JSON success status from the REST API (api.php), confirming that the POST request sent upon clicking **"Confirm Onsite Enrollment"** was successfully processed and the member data was inserted into the SQL database.

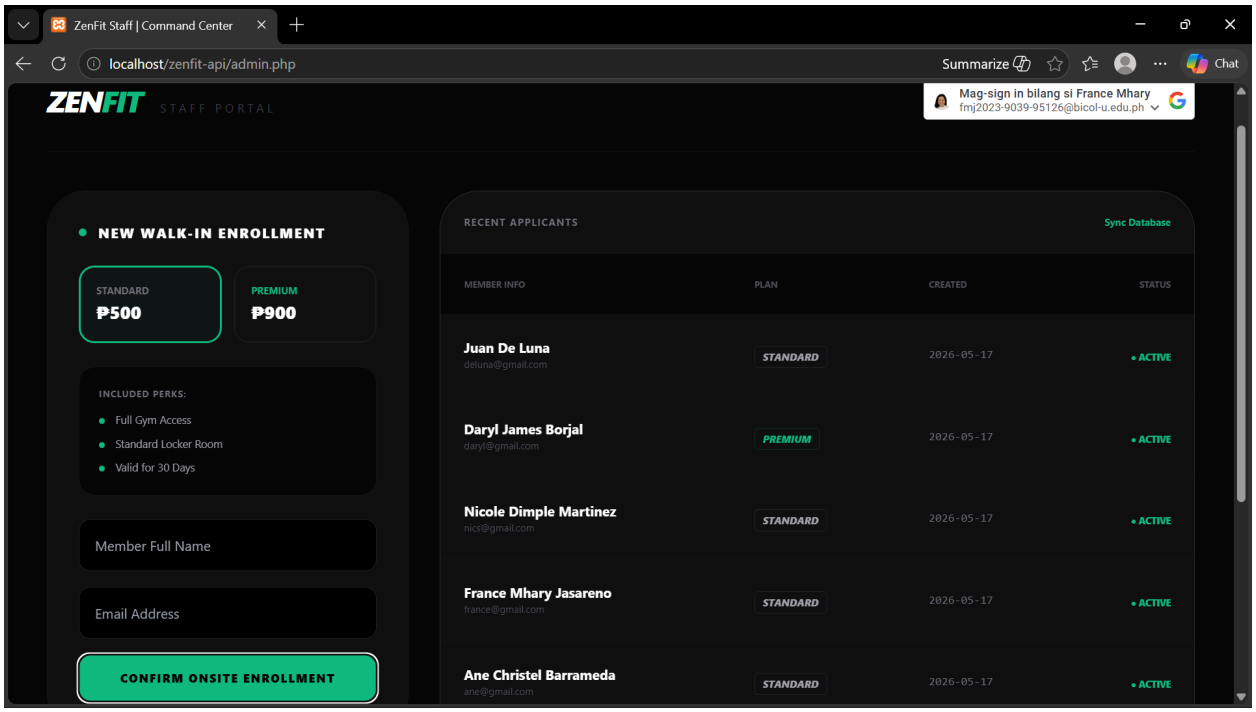

Figure 7: *Database Synchronization (Final State)*

The **"Juan De Luna"** entry appearing at the top of the **Recent Applicants** table confirms that the dashboard has automatically invoked the `loadDatabase()` function, which sends a fresh `GET` request to the API to retrieve the updated member list from the MySQL database and dynamically refresh the UI without a full page reload.